Foundations of Data Science

Question Catalogue

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1 Question

Find the SVD of the following matrix:

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$$

2 Question

Show that for any $A \in \mathbb{R}^{m \times n}$, the matrices $A^T A$ and AA^T possess the same nonzero eigenvalues.

3 Question

Given vectors

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad y = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix},$$

compute the distance between x and y using:

(a) $\langle x, y \rangle := x^T y$;

(b) $\langle x, y \rangle := x^T A y$, where $A := \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

4 Question

Consider the following bivariate distribution $p(x, y)$ of two discrete random variables X and Y.

$Y \backslash X$	x_1	x_2	x_3	x_4	x_5
y_1	0.01	0.02	0.03	0.1	0.1
y_2	0.05	0.1	0.05	0.07	0.2
y_3	0.1	0.05	0.03	0.05	0.04

Compute:

(a) The marginal distributions $p(x)$ and $p(y)$.

(b) The conditional distributions $p(x \mid Y = y_1)$ and $p(y \mid X = x_3)$.

5 Question

Consider a Gaussian random variable $x \sim \mathcal{N}(x \mid \mu_x, \Sigma_x)$, where $x \in \mathbb{R}^D$. Furthermore, we have

$$y = Ax + b + w,$$

where $y \in \mathbb{R}^E$, $A \in \mathbb{R}^{E \times D}$, $b \in \mathbb{R}^E$, and $w \sim \mathcal{N}(w \mid 0, Q)$ is independent Gaussian noise. "Independent" implies that x and w are independent random variables and that Q is diagonal.

(a) Write down the likelihood $p(y \mid x)$.

(b) The distribution $p(y) = \int p(y \mid x)\, p(x)\, dx$ is Gaussian. Compute the mean μ_y and the covariance Σ_y. Derive your result in detail.

(c) The random variable y is being transformed according to the measurement mapping

$$z = Cy + v,$$

where $z \in \mathbb{R}^F$, $C \in \mathbb{R}^{F \times E}$, and $v \sim \mathcal{N}(v \mid 0, R)$ is independent Gaussian (measurement) noise.

- Write down $p(z \mid y)$.
- Compute $p(z)$, i.e. the mean μ_z and the covariance Σ_z. Derive your result in detail.

6 Question

Let x and y be independent random variables with uniform distribution in $[-\frac{1}{2}, \frac{1}{2}]$. What is the expected value $E(x)$, $E(x^2)$, $E(x - y)$, $E(xy)$, and $E((x - y)^2)$?

7 Question

Let x be a random variable with probability density $\frac{1}{4}$ for $0 \leq x \leq 4$ and zero elsewhere.

1. Use Markov's inequality to bound the probability that $x \geq 3$.
2. Make use of $\text{prob}(|x| \geq a) = \text{prob}(x^2 \geq a^2)$ to get a tighter bound.
3. What is the bound using $\text{prob}(|x| \geq a) = \text{prob}(x^r \geq a^r)$?

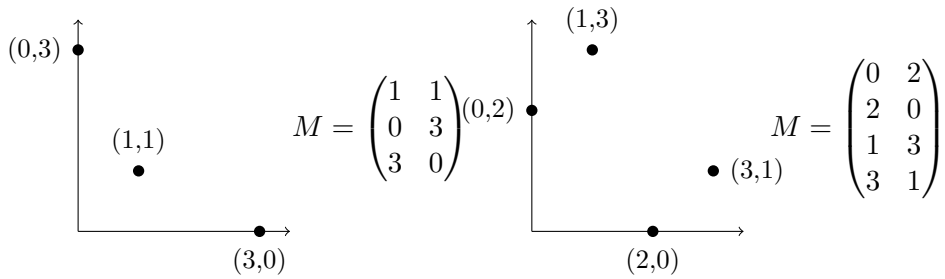
8 Question

Manually find the best-fit lines (not subspaces that must contain the origin) through the points in the sets below. Best fit means minimizing the perpendicular distance. Subtract the center of gravity of the points in the set from each of the points in the set and find the best-fit line for the resulting points. Does the best-fit line for the original data go through the origin?

1. $(4, 4)$ $(6, 2)$
2. $(4, 2)$ $(4, 4)$ $(6, 2)$ $(6, 4)$
3. $(3, 2.5)$ $(3, 5)$ $(5, 1)$ $(5, 3.5)$

9 Question

Manually find the left and right singular vectors, the singular values, and the SVD decomposition of the matrices in the following figure:



10 Question

Interpret the first right and left-singular vectors for the following *Term-Document Matrix*.

	D1	D2	D3	D4	D5
complexity	2		3	2	3
algorithm	3			4	4
entropy	1		2		
traffic		2			
network			3	1	4

(Documents → vector-space representation: term-document matrix.)

11 Question

Each of the following data sets consists of a subset of the d-dimensional 0/1 vectors labeled $+1$. The remaining 0/1 vectors are labeled -1. Which sets are linearly separable?

1. $\{010, 011, 100, 111\}$
2. $\{011, 100, 110, 111\}$
3. $\{0100, 0101, 0110, 1000, 1100, 1101, 1110, 1111\}$

12 Exercise Sheets – Questions

Sheet 0 Exercise 0.1 (Recall: Random variables)

(i) Consider the course's example: A player has a fair die D resulting in one of $\mathcal{D} := \{1, 2, 3, 4, 5, 6\}$ with probability $\frac{1}{6}$ each, i.e. $D \leftarrow \mathcal{D}$ uniformly, and a fair coin $C \leftarrow \{0, 1\}$ uniformly.

Actually, our player is a cheater and the coin merely tells whether he decides to cheat or not. In case the coin comes up heads, say that's encoded 1, he changes the die's outcome to 6. Denote the variable describing the faked die by F.

(a) Describe F as a function of C and D.

(b) Note that the pair (C, F) is again a random variable, namely with outputs in $\{0,1\}\times\mathcal{D}$. Write down a 2×6-table with its distribution.

(ii) Consider a continuous random variable U with outcomes in $[0,1] \subset \mathbb{R}$ with uniform density, i.e. $U \leftarrow [0,1]$ with density $p(x) = 1$ for $x \in [0,1]$ and $p(x) = 0$ otherwise. Determine the density of U^2, i.e. a function q such that

$$\operatorname{prob}\big(a < U^2 < b\big) = \int_a^b q(x)\,dx.$$

Sheet 0 Exercise 0.2 (Maximum of two dice)

Take two independent fair dice $D_i \leftarrow \mathcal{D} := \{1,2,3,4,5,6\}$ and consider the larger outcome $M := \max(D_0, D_1)$.

(i) Compute $\operatorname{prob}(M \leq a)$ for $a \in \mathbb{N}_{\leq 6}$.

(ii) Compute $\operatorname{prob}(M = a)$ for $a \in \mathbb{N}_{\leq 6}$.

(iii) Compute $\mathrm{E}(M)$.

(iv) Compute $\mathrm{E}(M^2)$ and derive $\operatorname{var}(M)$.

Sheet 1 Exercise 1.1 (Computer scientist's random variables)

Consider the following algorithm:

1. Throw a coin $C \leftarrow \{0,1\}$.
2. Choose $X \leftarrow [0,1]$ uniformly.
3. Roll a die $D \leftarrow \{1,2,3,4,5,6\}$.
4. **If** $C = 0$ **then**
5. Let $Z \leftarrow D + X$.
6. **Else**
7. Let $Z \leftarrow D - X$.
8. **Return** Z

The output Z of this algorithm is a random variable.

(i) Compute its expectation $\mathrm{E}(Z)$.

(ii) Compute its second moment $\mathrm{E}(Z^2)$.

(iii) Compute its variance $\operatorname{var}(Z)$.

Sheet 2 Exercise 2.2 (Annuli)

(i) Compute and estimate the volume of the $\frac{1}{100}$-annulus compared to the volume of the d-dimensional ball B^d.

(ii) Compute and estimate the volume of the $\frac{1}{\sqrt{d}}$-annulus compared to the volume of the d-dimensional ball B^d.

(iii) Compute and estimate the volume of the $\frac{1}{d^2}$-annulus compared to the volume of the d-dimensional ball B^d.

Sheet 4 Exercise 4.2 (Eigenvalues of graphs)

(i) What are the eigenvalues of (the adjacency matrices of) the graphs shown below? What does this say about using eigenvalues to determine whether two graphs are isomorphic?



Sheet 5 Exercise 5.3 (Eigenvalues and -vectors)

Take the matrix $A = \begin{bmatrix} 3 & 0 & 4 \\ 0 & -1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

(i) Compute eigenvalues and eigenvectors of A.

(iii) Compute $\|A\|_2$ and $\|A\|_F$ and compare.

13 Theoretical Questions – Self-Tests for Video Lectures

0000 Introduction

1000 High-dimensional space: Probabilities

1010 The Law of Large Numbers

1040 Properties of the unit ball

2000 Generating points uniformly at random from a ball

2020 Generating points uniformly at random from a ball, cont'd

2030 Gaussian

3020 Symmetric matrices

3030 Extremal properties of eigenvalues

3040 Eigenvalues of the sum of two symmetric matrices

4000 Best-fit subspaces and SVD: Introduction

4010 Best-fit subspaces and SVD: Singular vectors

4020 Best-fit subspaces and SVD: Singular value decomposition

4040 Best-fit subspaces and SVD: Left singular vectors

6020 Applications of SVD: Application 2, principal component analysis, collaborative filtering

7000 Machine Learning: Introduction

7010 Machine Learning: The perceptron algorithm

7020 Machine Learning: Kernel functions and non-linearly separable data

7030 Machine Learning: Generalizing to new data

7040 Machine Learning: VC-dimension: Definitions

14 Additional Questions from the Exercise Sheets

Exercise 5.4 (Eigenvalues of sums)

Consider the matrices

$$A = Q \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} Q^{-1}, \qquad B = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \text{with } Q = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$$

for some angle φ.

(i) List the eigenvalues of A and B.

(ii) Take $\varphi = 45° = \frac{\pi}{4}$. Compute the eigenvalues γ_i of $A + B$.

(iii) Compare the results with the theorem and the lemma about sums of symmetric matrices. *Hint:* In particular (but not exclusively!), discuss: Is the theorem correct? How large are the intervals? Would it make sense to swap A and B in the named results?

Exercise 9.3 (Kernels?)

For each of the following maps decide whether it is a kernel. Argue.

- $k(|x\rangle, |y\rangle) = \langle e_0 \,|\, x + y\rangle$ with the first standard unit vector $|e_0\rangle$.
- $k(|x\rangle, |y\rangle) = \langle x \,|\, y\rangle^5 + 17$.
- $k(|x\rangle, |y\rangle) = \cos\langle x \,|\, y\rangle$.
- $k(|x\rangle, |y\rangle) = \dfrac{\langle x \,|\, y\rangle}{\| |x\rangle \|_2 \, \| |y\rangle \|_2}$.

15 Clustering Exercises

CL1. Consider the following dataset in $\mathbb{R}^2$: $x_1 = (0, 0)$, $x_2 = (0, 1)$, $x_3 = (-1, 2)$, $x_4 = (2, 0)$, $x_5 = (3, 0)$, $x_6 = (4, -1)$.

(i) Compute the distance matrix using the Euclidean (L^2) distance.

(ii) Perform 1 iteration of k-means clustering with $k = 2$, using x_1 and x_6 as the initial cluster centers. What are the clusters and new centers?

CL2. Consider the following dataset in $\mathbb{R}^2$: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (2, 0)$, $x_4 = (-0.5, -1)$, $x_5 = (0.5, -1)$, $x_6 = (0, -1.5)$.

(i) Compute the distance matrix using the Manhattan (L^1) distance. Recall that the L^1 norm of a vector is given by $\|x\|_1 = \sum_i |x_i|$.

(ii) Perform agglomerative hierarchical clustering using single linkage and draw the resulting dendrogram (does not have to be to scale).

CL3. Let $A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$ with two clusterings:

- $C = \{C_0 = \{a_0, a_1, a_2\},\ C_1 = \{a_3, a_4, a_5\}\}$
- $C' = \{C_0 = \{a_0, a_3, a_4\},\ C_1 = \{a_1, a_2, a_5\}\}$

Compute the distance between the two clusterings $d(C, C')$.

CL4. Given data points in $\mathbb{R}^1$: $A = 1, 2, 3, 10, 11, 12$ and $k = 3$. Suppose that in k-means++ initialization, we have selected $c_0 = 2$ and $c_1 = 12$. What is the probability distribution for c_2?

16 VC Dimension and SVMs

VC1. Consider the class Ω of axis-parallel hypercubes (intervals, squares, cubes, ...) in $\Omega = \mathbb{R}^d$. This is more restricted than rectangles or hypercuboids! Namely, a hypercube is a hypercuboid with the additional requirement that all edge lengths are equal.

1. Show that for $d = 1$ we have $\text{VCdim}(\mathcal{H}) \geq 2$.
2. Show that for $d = 2$ we have $\text{VCdim}(\mathcal{H}) \geq 3$.
3. Show that for $d = 3$ we have $\text{VCdim}(\mathcal{H}) \geq 4$.

Hint: In each of these cases you have to specify one set of $2, 3, 5$ elements, respectively, and prove that it can be shattered by $\mathcal{H}$.

VC2. What is the VC dimension d of a support vector machine in $\mathbb{R}^2$? Give arguments why (for $d \geq \ldots$ and $d \leq \ldots$).
Hint: Draw pictures to support your claims.